

Puli VenkataReddy

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SUMMARY

DATA ENGINEER & ML PROFESSIONAL with 4+ years of experience designing scalable data pipelines and machine learning solutions. Proven ability to drive operational efficiency and revenue growth through advanced **data modeling**, **ETL automation**, and **cloud infrastructure**. Currently contributing to data engineering initiatives at **McKinsey & Company**, with a strong track record of reducing system downtime by 80% at **Dell** and improving revenue reconciliation by \$1.2M at **Airbnb**. Adept at mentoring teams and deploying production-ready AI systems.

PROFESSIONAL EXPERIENCE

NSM Solutions LLC – Data Analyst | Jan 2025 – Current | USA

- Accelerating onboarding and integration into the firm's data ecosystem, rapidly mastering proprietary data models and **ETL** frameworks to support client engagement teams.
- Initiating preliminary **data analysis** and **profiling** for a new **client project**, identifying initial data quality insights to shape the project's analytical approach.
- Collaborating with senior data engineers and architects to review and understand existing **data pipelines**, contributing to the design phase for upcoming **modernizations**.

HPE – Data Engineer / ML Engineer | June 2021 – May 2023 | India

- Designed and executed scalable ETL pipelines using **Python**, **PySpark**, and **Airflow**, processing over **500 million system logs daily**, improving ML model training speed and efficiency.
- Assembled supervised and unsupervised ML models using **scikit-learn**, **XGBoost**, and **TensorFlow** to predict server failures, reducing system downtime incidents from **15 per month to 3 per month**.
- Optimized **data storage and retrieval in AWS S3 and Redshift**, handling **multi-terabyte datasets** and reducing query execution times from **hours to under 10 minutes**.
- Orchestrated **feature engineering workflows** for predictive analytics, generating over **200 features across 50 datasets**, improving model accuracy and eliminating manual preprocessing tasks.
- Deployed **ML pipelines in Docker and Kubernetes clusters**, orchestrating over **100 scheduled jobs weekly**, including digitized model retraining and version control.
- Devised **real-time anomaly detection systems** using **Kafka streams** and **Spark Structured Streaming**, monitoring **2 million events per hour** and alerting engineering teams to critical system issues.
- Applied **data validation, logging, and monitoring frameworks** across 10+ projects using **Python**, **SQL**, and **CloudWatch**, improving reliability for downstream ML applications.
- Mentored a team of **3 junior engineers**, guiding Python scripting, ML modeling, and cloud data engineering practices, accelerating onboarding and shortening project delivery timelines.
- Created **interactive dashboards and reports** using **Tableau and Power BI**, visualizing ML model performance, operational KPIs, and system health metrics for cross-functional teams.
- Collaborated with **DevOps, Data Science, and IT teams** to integrate machine learning insights into production systems, enabling **predictive maintenance** that saved over **\$1M in operational costs annually**.

Wipro India Ltd – Data Analyst | May 2020 – May 2021 | India

- Engineered **SQL queries and Python scripts** to analyze **120 million booking records**, identifying anomalies and improving revenue reconciliation by **\$1.2M annually**.
- Constructed **interactive dashboards in Tableau and Power BI** for 15+ teams, visualizing **customer engagement trends** and reducing manual reporting time from **10 hours to 2 hours per week**.
- Conducted **data validation and anomaly detection** across multi-source datasets, improving quality for predictive modeling pipelines used by **3 internal ML teams**.
- Enforced **simplified ETL workflows** in Python and Airflow, processing over **50 GB of daily data**, decreasing report generation lag from **48 hours to 6 hours**.
- Performed **statistical analysis and hypothesis testing** to optimize pricing strategies across 20+ markets, generating an additional **\$500K in quarterly revenue**.
- Collaborated with **product, marketing, and engineering teams** to design data-driven KPIs, tracking retention and engagement for **10M+ active users**, enhancing decision-making for product improvements.
- Analyzed **customer reviews and feedback datasets** using **Python NLP libraries (NLTK, spaCy)** to extract insights on user satisfaction, identifying top 5 recurring issues to prioritize in product development.
- Monitored **A/B testing outcomes** for platform feature rollouts, applying **R and Python analytics** to measure user behavior and guide strategic updates across 50+ experimental cohorts.

HCL – Data Analyst Intern | May 2019 – April 2020 | India

- Streamlined **data extraction and transformation** workflows using **Alteryx and Excel VBA**, processing **30M+ transactional records**, reducing ETL runtime from **12 hours to 3 hours**.
- Generated **custom dashboards in Power BI and Tableau**, monitoring client revenue, bookings, and operational KPIs across 5 projects, enabling management to identify **\$800K in cost-saving opportunities**.

- Conducted **statistical and correlation analysis** on multi-source datasets using **R and SPSS**, identifying trends that informed **6 quarterly business strategy adjustments**.
- Built **predictive inventory and booking models** with **Prophet and Excel Solver**, reducing decision-making cycles by **7 days** for client supply chain and operational planning.
- Integrated datasets from **MySQL, Oracle, and Google BigQuery** into centralized repositories, supporting **10+ analytical models** and enabling cross-project reporting.
- Launched **data quality monitoring frameworks** with **Dataedo and dbt**, detecting anomalies and reducing inaccurate reporting incidents from **14 to 2 per quarter**.
- Performed **ad-hoc financial and operational analysis** using **Power Query and Excel PivotTables**, producing insights that guided **resource allocation for 5 client programs**.

TECHNICAL SKILLS

- **Programming & ML Tools:** Python (NumPy, Pandas, Matplotlib, Seaborn), R, SQL, PySpark, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, FastAPI
- **Data Engineering & Storage:** Apache Kafka, Apache Airflow, Hadoop, Spark, MongoDB, PostgreSQL, Snowflake, Redshift, Delta Lake
- **Cloud & Platforms:** AWS (S3, EMR, Lambda, SageMaker), Azure (ML Studio, Cognitive Services, Data Lake), GCP (BigQuery, Vertex AI), Databricks
- **AI & GenAI Tools:** FAISS, OpenAI APIs, Vector Databases (Pinecone, Weaviate), Retrieval-Augmented Generation (RAG), Generative AI, Explainable AI (XAI)
- **Data Visualization & Business Intelligence:** Tableau, Power BI, QuickSight, Excel (PivotTables, VLOOKUP, Dashboards), KPI Reporting, Matplotlib, Seaborn
- **Data Analysis & Statistical Skills:** Data Cleaning, Data Transformation, Hypothesis Testing, A/B Testing, Regression Analysis, Time Series Forecasting, Predictive Modeling
- **Model Deployment & MLOps:** MLflow, TFX, Kubeflow, Model Registry, Model Monitoring & Drift Detection, Endpoint Optimization, CI/CD Pipelines
- **Core Strengths:** Data Pipelines, Model Deployment, Real-Time & Streaming Analytics, ETL Automation, Dashboard & Report Development, Data-Driven Decision Making

EDUCATION

Elmhurst University | Chicago, IL, USA

Master of Science (Computer Information Technology) | 2023 – 2025

PROJECTS

Production-Ready LLM RAG Agent for Domain Q&A

- Innovated a **Retrieval-Augmented Generation (RAG) agent** leveraging large language models to answer domain-specific queries with high accuracy.
- Integrated **vector databases** to store and retrieve embeddings, enabling efficient semantic search across large document corpora.
- Deployed a **production-ready pipeline** with standardized query handling, improving response time and delivering scalable AI-powered knowledge solutions.

Automated Model Monitoring & Continuous Retraining Platform (MLOps)

- Developed a **systematized model monitoring system** to track performance metrics, data drift, and prediction accuracy in real time.
- Implemented **continuous retraining pipelines** using CI/CD frameworks to update machine learning models dynamically based on new data.
- Enabled **scalable and reliable deployment** of ML models, reducing manual intervention and ensuring consistent prediction quality across production environments.